carry the three edges emenating from that vertex cyclically into themselves. Thus $|X_{\text{3-cycle}}| = 2 \cdot 2$. The element (1, 2)(3, 4) of G carries the edge joining vertex 1 to vertex 2 and the edge joining vertex 3 to vertex 4 into themselves, swaps the edge joining vertices 1 and 3 with the one joining vertices 2 and 4, and swaps the edge joining vertices 1 and 4 with the one joining vertices 2 and 3. Thus we see that $|X_{(1,2)(3,4)}| = 2 \cdot 2 \cdot 2 \cdot 2$, and of course the analogous count can be made for the group elements (1, 3)(2, 4) and (1, 4)(2, 3). Thus we obtain

$$|X_\iota| = 2^6, \quad |X_{\text{3-cycle}}| = 2^2, \quad \text{and} \quad |X_{\text{other type}}| = 2^4.$$

Consequently $\sum_{g \in G} |X_g| = 2^6 + 8 \cdot 2^2 + 3 \cdot 2^4 = 144$. The number of distinguishable blocks is thus $(1/12)(144) = 12$.

9. a. The group G of rigid motions of the prism has order 8, four positions leaving the end faces in the same position and four positions with the end faces swapped. There are $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ ways of painting the faces different colors. We let X be the set of these 6! possibilities. Then $|X_\iota| = 6!$ and $|X_{\text{other}}| = 0$. Thus there are $(1/8)(6!) = 6 \cdot 5 \cdot 3 = 90$ distinguishable painted prisms using six different colors.

b. This time the set X of possible ways of painting the prism has 6^6 elements. We have
$|X_\iota| = 6^6$ where ι is the identity element,
$|X_{\text{same ends, rotate 90° or 270°}}| = 6 \cdot 6 \cdot 6$,
$|X_{\text{same ends, rotate 180°}}| = 6 \cdot 6 \cdot 6 \cdot 6$,
$|X_{\text{swap ends, keeping top face on top}}| = 6 \cdot 6 \cdot 6 \cdot 6$,
$|X_{\text{swap ends, as above, rotate 90° or 270°}}| = 6 \cdot 6$,
$|X_{\text{swap ends, as above, rotate 180°}}| = 6 \cdot 6 \cdot 6$.
Thus $\sum_{g \in G} |X_g| = 6^6 + 2 \cdot 6^3 + 6^4 + 6^4 + 2 \cdot 6^2 + 6^3 = 6^2(6^4 + 92)$. The number of distinguishable blocks is thus $(1/8)[6^2(6^4 + 82)] = 6,246$.

18. Rings and Fields

1. 0 **2.** 16 **3.** 1 **4.** 22 **5.** (1, 6) **6.** (2, 2)

7. Yes, $n\mathbb{Z}$ for $n \in \mathbb{Z}^+$ is a commutative ring, but without unity unless $n = 1$, and is not a field.

8. No, $\mathbb{Z}^+$ is not a ring; there is no identity for addition.

9. Yes, $\mathbb{Z} \times \mathbb{Z}$ is a commutative ring with unit (1, 1), but is not a field because (2, 0) has no multiplicative inverse.

10. Yes, $2\mathbb{Z} \times \mathbb{Z}$ is a commutative ring, but without unity, and is not a field.

11. Yes, $\{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$ is a commutative ring with unity, but is not a field because 2 has no multiplicative inverse.

12. Yes, $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is a commutative ring with unity and is a field because

$$\frac{1}{a + b\sqrt{2}} = \frac{1}{a + b\sqrt{2}} \cdot \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a}{a^2 - 2b^2} + \frac{-b}{a^2 - 2b^2}\sqrt{2}.$$

13. No, $\mathbb{R}i$ is not closed under multiplication. **14.** The units in $\mathbb{Z}$ are 1 and -1.

15. The units in $\mathbb{Z} \times \mathbb{Z}$ are (1, 1), (1, -1), (-1, 1), and (-1, -1).

16. The units in $\mathbb{Z}_5$ are 1, 2, 3, and 4 because $1 \cdot 1 = 2 \cdot 3 = 4 \cdot 4 = 1$.

17. All nonzero elements of $\mathbb{Q}$ are units.

18. The units in $\mathbb{Z} \times (Q) \times \mathbb{Z}$ are $(1, q, 1), (-1, q, 1), (1, q, -1)$ and $(-1, q, -1)$ for any nonzero $q \in \mathbb{Q}$.

19. The units in $\mathbb{Z}_4$ are 1 and 3; $1 \cdot 1 = 3 \cdot 3 = 1$.

20. **a.** Each of the four entries in the matrix can be either of two elements, so there are $2^4 = 16$ matrices in $M_2(\mathbb{Z}_2)$.

b. $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = I_2$, and $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ are all their own inverse. These six matrices are the units.

21. (See the answer in the text.)

22. Because $\det(A + B)$ need not equal $\det(A) + \det(B)$, we see that det is not a ring homomorphism. For example, $\det(I_n + I_n) = 2^n$ but $\det(I_n) + \det(I_n) = 1 + 1 = 2$.

23. Let $\phi : \mathbb{Z} \to \mathbb{Z}$ be a ring homomorphism. Because $1^2 = 1$, we see that $\phi(1)$ must be an integer whose square is itself, namely either 0 or 1. If $\phi(1) = 1$ then $\phi(n) = \phi(n \cdot 1) = n$, so ϕ is the identity map of $\mathbb{Z}$ onto itself which is a homomorphism. If $\phi(1) = 0$, then $\phi(n) = \phi(n \cdot 1) = 0$, so ϕ maps everthing onto 0, which also yields a homomorphism.

24. As in the preceding solution, we see that for a ring homomorphism $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$, we must have $\phi(1)^2 = \phi(1^2) = \phi(1)$. The only elements of $\mathbb{Z} \times \mathbb{Z}$ that are their own squares are (0, 0), (1, 0), (0, 1), and (1, 1). Thus the possibilities for ϕ are $\phi_1(n) = (0, 0), \phi_2(n) = (n, 0), \phi_3(n) = (0, n)$, and $\phi_4(n) = (n, n)$. It is easily checked that these four maps are ring homomorphisms.

25. Because both (1, 0) and (0, 1) are their own squares, their images under a ring homomorphism $\phi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ must also have this property, and thus must be either 0 or 1. Because (1, 0) and (0, 1) generate $\mathbb{Z} \times \mathbb{Z}$ as an additive group, this determines the possible values of the homomorphism on $(n, m) \in \mathbb{Z} \times \mathbb{Z}$. Thus the possibilities are given by $\phi_1(n, m) = 0, \phi_2(n, m) = n, \phi_3(n, m) = m$, and $\phi_4(n, m) = n + m$. It is easily checked that ϕ_1, ϕ_2, and ϕ_3 are homomorpisms. However, ϕ_4 is not a homomorphism because $n + m = \phi_4(n, m) = \phi_4((1, 1)(n, m)) \neq \phi_4(1, 1)\phi_4(n, m) = (1 + 1)(n + m) = 2(n + m)$.

26. As in Exercise 25, we see that the images of additive generators (1, 0, 0), (0,1, 0), and (0, 0, 1) under a ring homomorphism $\phi : \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ can only be 0 and 1. As shown in the argument for ϕ_4 in that exercise, mapping more than one of these generators into 1 will not give a ring homomorphism. Thus either they are all mapped into 0 giving the trivial homomorphism, or we have a projection homomorphism where one of the generators is mapped into 1 and the other two are mapped into 0. Thus there are four such ring homomorphisms ϕ.

27. (See the answer in the text)

28. We have $x^2 + x - 6 = (x + 3)(x - 2)$. In $\mathbb{Z}_{14}$, it is possible to have a product of two nonzero elements be 0. Trying all elements x from -6 to 7 in $\mathbb{Z}_{14}$ to see if $(x+3)(x-2)$ is zero, we find that this happens for $x = -5, -3, 2$, and 4. Thus the elements 2, 4, 9, and 11 in $\mathbb{Z}_{14}$ are solutions of the quadratic equation.

29. The definition is incorrect. Insert the word "commutative" before either "ring" or "group".

A **field** F is a commutative ring with nonzero unity such that the set of nonzero elements of F is a group under multiplication.

30. The definition is incorrect. We have not defined any concept of magnitude for elements of a ring.

A **unit** in a ring with nonzero unity is an element that has a multiplicative inverse.

31. In the ring $\mathbb{Z}_6$, we have $2 \cdot 3 = 0$, so we take $a = 2$ and $b = 3$.

32. $\mathbb{Z} \times \mathbb{Z}$ has unity (1, 1); however the subring $\mathbb{Z} \times \{0\}$ has unity (1, 0). Also, $\mathbb{Z}_6$ has unity 1 while the subring $\{0, 2, 4\}$ has unity 4, and the subring $\{0, 3\}$ has unity 3.

33. T F F F T F T T T T

34. Let $f, g, h \in F$. Now $[(fg)h](x) = [(fg)(x)]h(x) = [f(x)g(x)]h(x)$. Because multiplication in $\mathbb{R}$ is associative, we continue with $[f(x)g(x)]h(x) = f(x)[g(x)h(x)] = f(x)[(gh)(x)] = [f(gh)](x)$. Thus $(fg)h$ and $f(gh)$ have the same value on each $x \in \mathbb{R}$, so they are the same function and axiom 2 holds. For axiom 3, we use the distributive laws in $\mathbb{R}$ and we have $[f(g+h)](x) = f(x)[(g+h)(x)] = f(x)[g(x)+h(x)] = f(x)g(x)+g(x)h(x) = (fg)(x)+(fh)(x) = (fg+fh)(x)$ so $f(g+h)$ and $fg+fh$ are the same function and the left distributive law holds. The right distributive law is proved similarly.

35. For $f, g \in F$, we have $\phi_a(f+g) = (f+g)(a) = f(a) + g(a) = \phi_a(f) + \phi_a(g)$. Turning to the multiplication, we have $\phi_a(fg) = (fg)(a) = f(a)g(a) = \phi_a(f)\phi_a(g)$. Thus ϕ_a is a homomorphism.

36. We need check only the multiplicative property.

Reflexive: The identity map ι of a ring R into itself satisfies $\iota(ab) = ab = \iota(a)\iota(b)$, so the reflexive property is satisfied.

Symmetric: Let $\phi : R \to R'$ be an isomorphism. We know from group theory that $\phi^{-1} : R' \to R$ is an isomorphism of the additive group of R' with the additive group of R. For $\phi(a), \phi(b) \in R'$, we have $\phi^{-1}(\phi(a)\phi(b)) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(\phi(a))\phi^{-1}(\phi(b))$.

Transitive: Let $\phi : R \to R'$ and $\psi : R' \to R''$ be ring isomorphisms. Exercise 27 of Section 3 shows that $\psi\phi$ is an isomorphism of both the additive binary structure and the multiplicative binary stucture. Thus $\psi\phi$ is again a ring isomorphism.

37. Let $u, v \in U$. Then there exists $s, t \in R$ such that $us = su = 1$ and $vt = tv = 1$. These equations show that s and t are also units in U. Then $(ts)(uv) = t(su)v = t1v = tv = 1$ and $(uv)(ts) = u(vt)s = u1s = 1$, so uv is again a unit and U is closed under multiplication. Of course multiplication in U is associative because multiplication in R is associative. The equation $(1)(1) = 1$ shows that 1 is a unit. We showed above that a unit u in U has a multiplicative inverse s in U. Thus U is a group under multiplication.

38. Now $(a+b)(a-b) = a^2 + ba - ab - b^2$ is equal to $a^2 - b^2$ if and only if $ba - ab = 0$, that is, if and only if $ba = ab$. But $ba = ab$ for all $a, b \in R$ if and only if R is commutative.

39. We need only check the second and third ring axioms. For axiom 2, we have $(ab)c = 0c = 0 = a0 = a(bc)$. For axiom 3, we have $a(b+c) = 0 = 0 + 0 = ab + ac$, and $(a+b)c = 0 = 0 + 0 = ac + bd$.

40. If $\phi : 2\mathbb{Z} \to 3\mathbb{Z}$ is an isomorphism, then by group theory for the additive groups we know that either $\phi(2) = 3$ or $\phi(2) = -3$, so that either $\phi(2n) = 3n$ or $\phi(2n) = -3n$. Suppose that $\phi(2n) = 3n$. Then

$\phi(4) = 6$ while $\phi(2)\phi(2) = (3)(3) = 9$. Thus $\phi(2n) = 3n$ does not give an isomorphism, and a similar computation shows that $\phi(2n) = -3n$ does not give an isomorphism either.

$\mathbb{R}$ and $\mathbb{C}$ are not isomorphic because every element in the field $\mathbb{C}$ is a square while -1 is not a square in $\mathbb{R}$.

41. In a commutative ring, we have $(a+b)^2 = a^2+ab+ba+b^2 = a^2+ab+ab+b^2 = a^2+2\cdot ab+b^2$. Now the binomial theorem simply counts the number of each type of product a^ib^{n-i} appearing in $(a+b)^n$. As long as our ring is commutative, every summand of $(a+b)^n$ can be written as a product of factors a and b with all the factors a written first, so the usual binomial expansion is valid in a commutative ring.

In $\mathbb{Z}_p$, the coefficient $\binom{p}{i}$ of a^ib^{p-i} in the expansion of $(a+b)^p$ is a multiple of p if $1 \le i \le p-1$. Because $p \cdot a = 0$ for all $a \in \mathbb{Z}_p$, we see that the only nonzero terms in the expansion are those corresponding to $i = 0$ and $i = p$, namely b^p and a^p.

42. Let F be a field, and suppose that $u^2 = u$ for nonzero $u \in F$. Multiplying by u^{-1}, we find that $u = 1$. This shows that 0 and 1 are the only solutions of the equation $x^2 = x$ in a field. Now let K be a subfield of F. The unity of K satisfies the equation $x^2 = x$ in K, and hence also in F, and thus must be the unity 1 of F.

43. Let u be a unit in a ring R. Suppose that $su = us = 1$ and $tu = ut = 1$. Then $s = s1 = s(ut) = (su)t = 1t = t$. Thus the inverse of a unit is unique.

44. **a.** If $a^2 = a$ and $b^2 = b$ and if the ring is commutative, then $(ab)^2 = abab = aabb = a^2b^2 = ab$, showing that the idempotents are closed under multiplication.

b. By trying all elments, we find that the idempotents in $\mathbb{Z}_6$ are 0, 1, 3, and 4 while the idempotents in $\mathbb{Z}_{12}$ are 0, 1, 4, and 9. Thus the idempotents in $\mathbb{Z}_6 \times \mathbb{Z}_{12}$ are

$$\begin{array}{llll} (0,\,0) & (0,\,1) & (0,\,4), & (0,\,9) \\ (1,\,0) & (1,\,1) & (1,\,4), & (1,\,9) \\ (3,\,0) & (3,\,1) & (3,\,4), & (3,\,9) \\ (4,\,0) & (4,\,1) & (4,\,4), & (4,\,9). \end{array}$$

45. We have

$$\begin{aligned} P^2 &= [A(A^TA)^{-1}A^T][A(A^TA)^{-1}A^T] = A[(A^TA)^{-1}(A^TA)](A^TA)^{-1}A \\ &= AI_n(A^TA)^{-1}A^T = A(A^TA)^{-1}A^T = P. \end{aligned}$$

46. As explained in the answer to Exercise 41, the binomial expansion is valid in a commutative ring. Suppose that $a^n = 0$ and $b^m = 0$ in R. Now $(a+b)^{m+n}$ is a sum of terms containing as a factor a^ib^{m+n-i} for $0 \le i \le m+n$. If $i \ge n$, then $a^i = 0$ so each term with a factor a^ib^{m+n-i} is zero. On the other hand, if $i < n$, then $m+n-i > m$ so $b^{m+n-i} = 0$ and each term with a factor a^ib^{m+n-i} is zero. Thus $(a+b)^{m+n} = 0$, so $a+b$ is nilpotent.

47. If R has no nonzero nilpotent element, then the only solution of $x^2 = 0$ is 0, for any nonzero solution would be a nilpotent element. Conversely, suppose that the only solution of $x^2 = 0$ is 0, and suppose that $a \neq 0$ is nilpotent. Let n be the smallest positive integer such that $a^n = 0$. If n is even, then $a^{n/2} \neq 0$ but $(a^{n/2})^2 = a^n = 0$ so $a^{n/2}$ is a nonzero solution of $x^2 = 0$, contrary to assumption. If n is odd, then $(a^{(n+1)/2})^2 = a^{n+1} = a^na = 0a = 0$ so $a^{(n+1)/2}$ is a nonzero solution of $x^2 = 0$, contrary to assumption. Thus R has no nonzero nilpotent elements.

48. It is clear that if S is a subring of R, then all three of the conditions must hold. Conversely, suppose the conditions hold. The first two conditions and Exercise 45 of Section 5 show that $\langle S, + \rangle$ is an additive group. The final condition shows that multiplication is closed on S. Of course the associative and distributive laws hold for elements of S, because they actually hold for all elements in R. Thus S is a subring of R.

49. a. Let R be a ring and let $H_i \leq R$ for $i \in I$. Theorem 7.4 shows that $H = \bigcap_{i \in I} H_i$ is an additive group. Let $a, b \in H$. Then $a, b \in H_i$ for $i \in I$, so $ab \in H_i$ for $i \in I$ because H_i is a subring of R. Therefore $ab \in H$ so H is closed under multiplication. Clearly the associative and distributive laws hold for elements from H, because they actually hold for all elements in R. Thus H is a subring of R.

b. Let F be a field, and let $K_i \leq F$ for $i \in I$. Part(**a**) shows that $K = \bigcap_{i \in I} K_i$ is a ring. Let $a \in K, a \neq 0$. Then $a \in K_i$ for $i \in I$ so $a^{-1} \in K_i$ for $i \in I$ because Exercises 42 and 43 show that the unity in each K_i is the same as in F and that inverses are unique. Therefore $a^{-1} \in K$. Of course multiplication in K is commutative because multiplication in F is commutative. Therefore K is a subfield of F.

50. We show that I_a satisfies the conditions of Exercise 48. Because $a0 = 0$ we see that $0 \in I_a$. Let $c, d \in I_a$. Then $ac = ad = 0$ so $a(c - d) = ac - ad = 0 - 0 = 0$; thus $(c - d) \in I_a$. Also $a(cd) = (ac)d = 0d = 0$ so $cd \in I_a$. This completes the check of the properties in Exercise 48.

51. Clearly a^n is in every subring containing a, so R_a contains a^n for every positive integer n. Thus $\langle R_a, + \rangle$ contains the additive group G generated by $S = \{a^n \mid n \in \mathbb{Z}^+\}$. We claim that $G = R_a$. We need only show that G is closed under multiplication. Now G consists of zero and all finite sums of terms of the form a^n or $-a^m$. By the distributive laws, the product of two elements that are finite sums of positive powers and inverses of postive powers of a can again be written as such a sum, and is thus again in G. Therefore G is actually a subring containing a and contained in R_a so we must have $G = R_a$.

52. Example 18.15 shows that the map $\phi : \mathbb{Z}_{rs} \to \mathbb{Z}_r \times \mathbb{Z}_s$ where $\phi(a) = a \cdot (1, 1)$ is an isomorphism. Let $b = \phi^{-1}(m, n)$. Computing $b \cdot (1, 1)$ by components, we see that $1 + 1 + \cdots + 1$ for b summands yields m in $\mathbb{Z}_r$ and yields n in $\mathbb{Z}_s$. Thus, viewing b as an integer in $\mathbb{Z}$, we see that $b \equiv m \pmod{r}$ and $b \equiv n \pmod{s}$.

53. a. Statement: Let $b_1, b_2, \cdots, b_n$ be integers such that $\gcd(b_i, b_j) = 1$ for $i \neq j$. Then $\mathbb{Z}_{b_1 b_2 \cdots b_n}$ is isomorphic to $\mathbb{Z}_{b_1} \times \mathbb{Z}_{b_2} \times \cdots \mathbb{Z}_{b_n}$ with an isomorphism ϕ where $\phi(1) = (1, 1, \cdots, 1)$.

Proof: By the hypothesis that $\gcd(b_i, b_j) = 1$ for $i \neq j$, we know that the image group is cyclic and that $(1, 1, \cdots, 1)$ generates the group. Because the domain group is cyclic generated by 1, we know that ϕ is an additive group isomorphism. It remains to show that $\phi(ms) = \phi(m)\phi(s)$ for m and s in the domain group. This follows from the fact that the ith component of $\phi(ms)$ in the image group is $(ms) \cdot 1$ which is equal to the product of m summands of 1 times s summands of 1 by the distributive laws in a ring.

b. Let $c = \phi^{-1}(a_1, a_2, \cdots, a_n)$ where ϕ be the isomorphism in Part(**a**). Computing $\phi(c) = \phi(c \cdot 1)$ in its ith component, we see that $1 + 1 + \cdots + 1$ for c summands in the ring $\mathbb{Z}_{b_i}$ yields a_i. Viewing c as an integer, this means that $c \equiv a_i \pmod{\ b_i}$.

54. Note that $a0 = 0$ for all $a \in S$ follows from the distributive laws, so associativity of multiplication for products containing a factor 0 holds, and associativity in the group $\langle S^*, \cdot \rangle$ takes care of associativity for other products. All of the other axioms needed to verify that S is a division ring follow at once from the two given group statements and the given distribtive laws, except for the commutativity of addition.